ATI COURSE PROJECT 2026 · TRACK 2 · MIDTERM

ViGoEmotions

Fine-grained multi-label emotion recognition for Vietnamese social-media comments

[Team name] · [Member 1], [Member 2], [Member 3]

[Presentation date]

THE TASK

One comment can carry several emotions at once

ILLUSTRATIVE COMMENT (NOT FROM THE DATASET)

“Đọc xong muốn khóc luôn 😭 thương ông bà quá”

sadness | love | caring

Three labels at once: the model makes 28 independent yes/no decisions.

28 independent yes/no decisions

27 GoEmotions categories + neutral; one sigmoid per label, never a softmax.

Headline metrics: Macro-F1 and Micro-F1

Exact-match accuracy punishes any partial miss across 28 labels, so it is reported but not ranked on.

Informal Vietnamese

Teencode, slang, emoji and emoticons carry much of the emotional signal.

DATASET AND ACCESS PROTOCOL

20,664 comments from six platforms

20,664
comments (16,531 / 2,066 / 2,067)

28
labels: 27 emotions + neutral

1.91
labels per comment on average

1,485
distinct label combinations

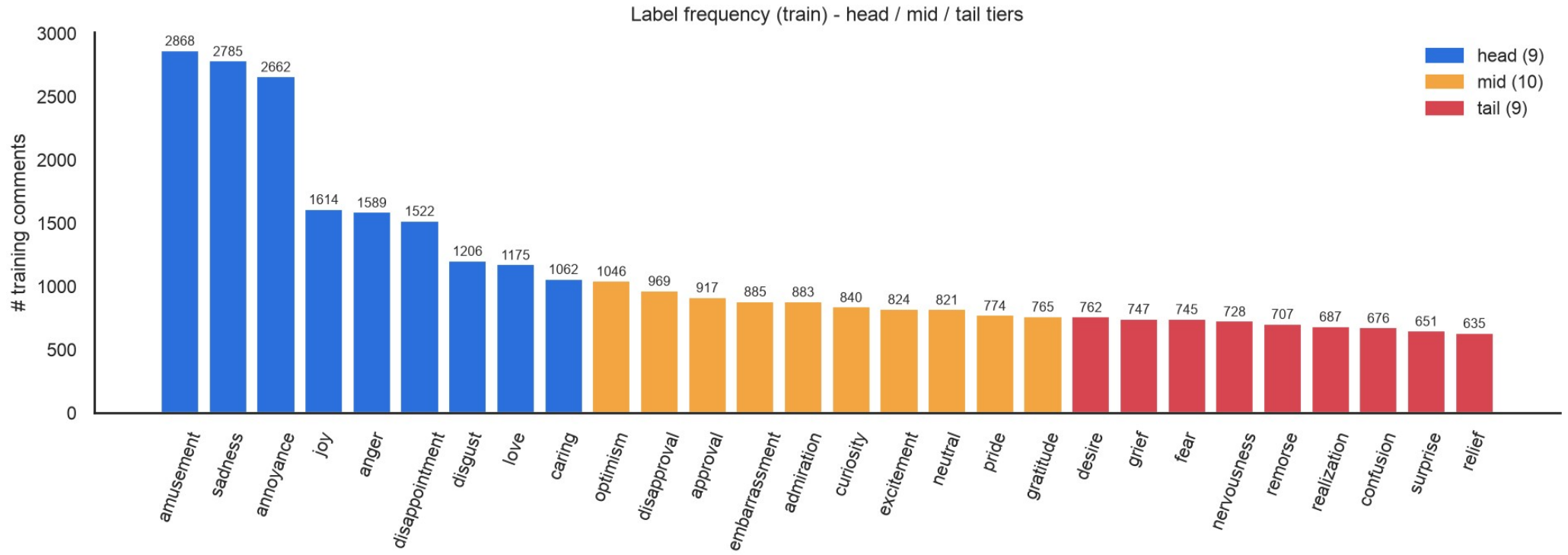
Source (from sample id)	Comments	Share
Facebook (numeric id, incl. UIT-VSMEC)	13,444	65.1%
TikTok	4,508	21.8%
YouTube	1,960	9.5%
X · Reddit · Threads	752	3.6%

Licence and access

Gated Hugging Face dataset uitnlp/vigoemotions (Hung et al., EACL 2026). Non-commercial research only, no redistribution: our repository holds code, split ids, aggregates and probabilities — never raw comments.

CLASS IMBALANCE

Three head labels dominate; the imbalance is only 4.5×



Head (9)

1,062–2,868 training positives: amusement, sadness, annoyance lead.

Mid (10)

765–1,046, including neutral.

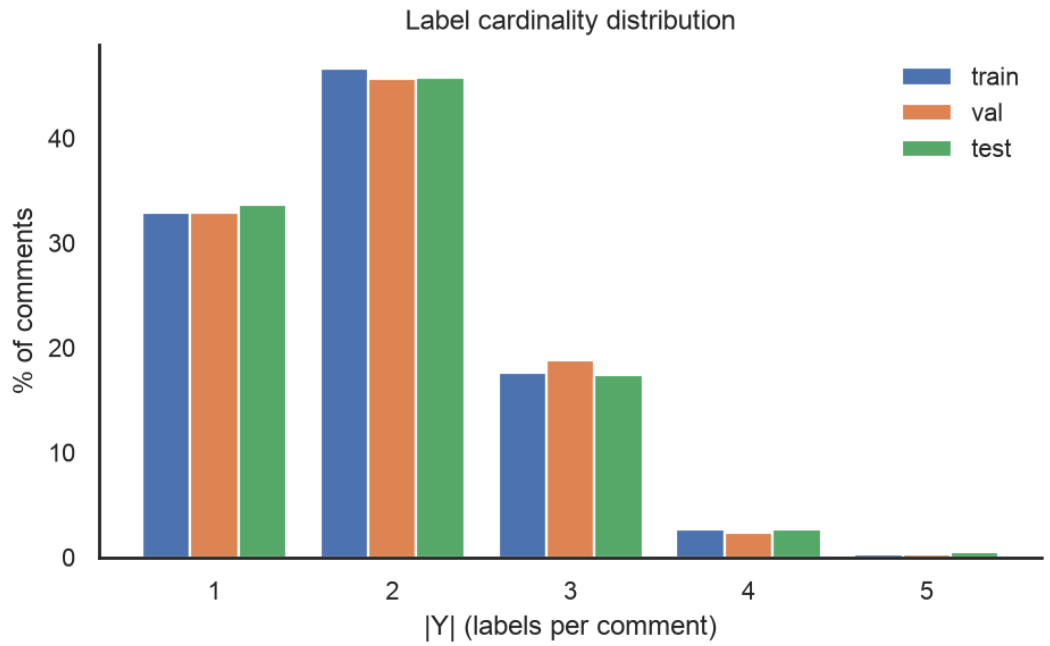
Tail (9)

635–762: desire … surprise, relief.

Tier rule fixed before any evaluation: tertiles of TRAINING positive counts · results/tables/label_tiers.csv

LABEL CARDINALITY AND CO-OCCURRENCE

Two labels per comment is the norm, not the exception



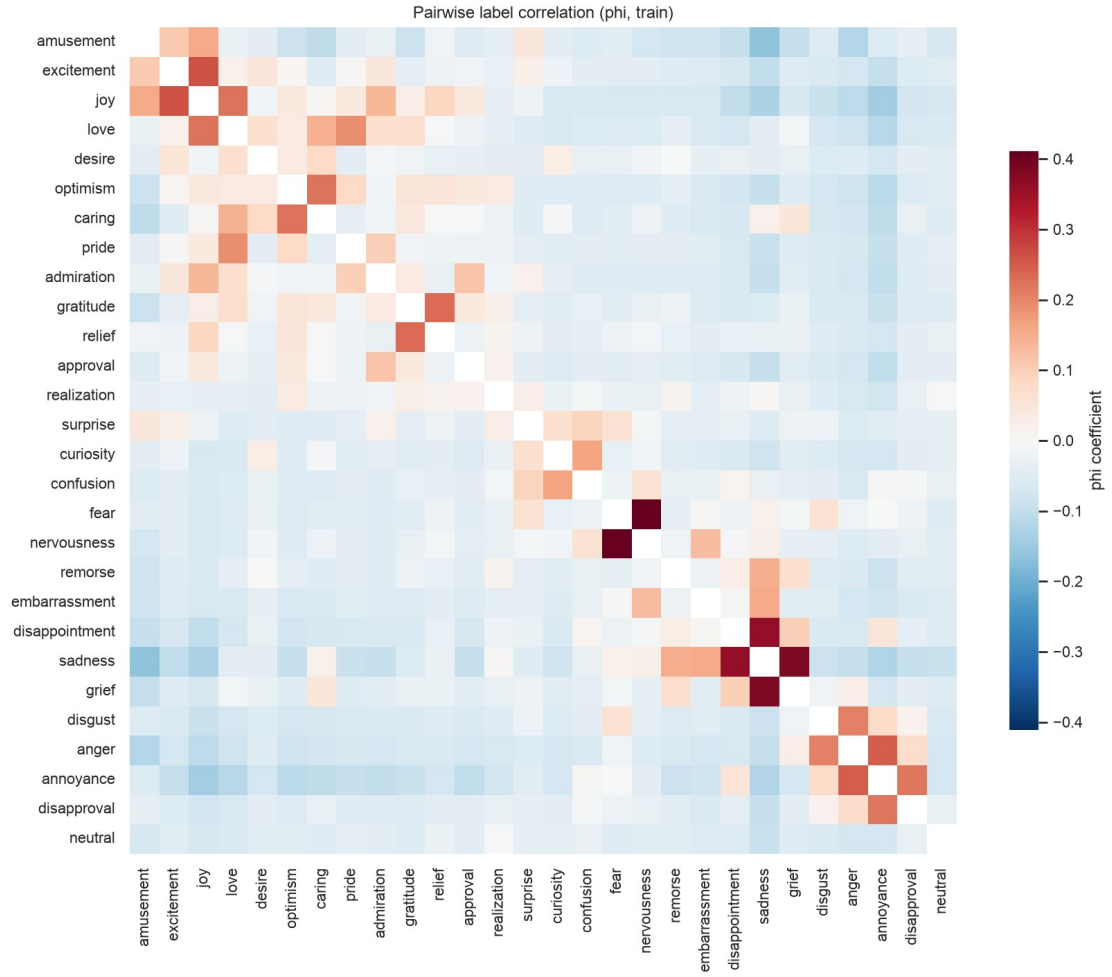
|Y| = 1: 32.9% · 2: 46.6% · 3: 17.6% · ≥4: 2.9% (train). Only 3.5% are neutral-only.

Most frequent pairs (train)

Pair	Count	P(b|a)	phi
disappointment + sadness	904	0.59	0.36
anger + annoyance	699	0.44	0.25
sadness + grief	620	0.22	0.38
amusement + joy	568	0.20	0.15
annoyance + disapproval	469	0.18	0.22
joy + love	395	0.24	0.22
disgust + anger	379	0.31	0.21
embarrassment + sadness	366	0.41	0.16

83% of grief comments are also labelled sadness.

Computed on training labels only · cardinality_summary.csv, top_pairs.csv



PAIRWISE CORRELATION (PHI)

Emotions cluster into families

Sadness family

sadness – grief 0.38, disappointment – sadness 0.36

Anxiety

fear – nervousness 0.41 (strongest)

Positive

excitement – joy, joy – love, optimism – caring, gratitude – relief

Hostile

anger – annoyance – disapproval, disgust – anger

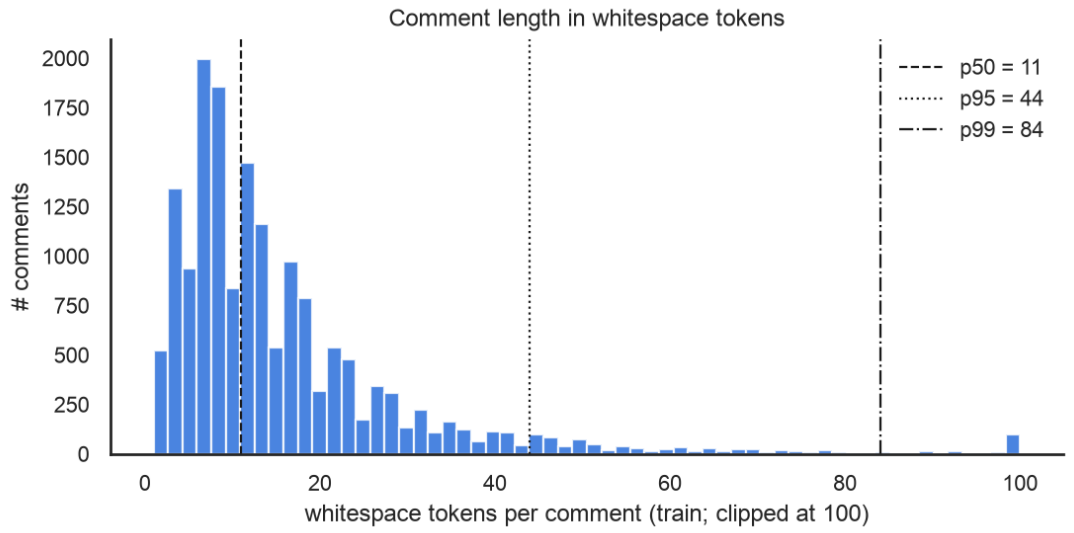
Opposites

amusement is negatively correlated with sadness

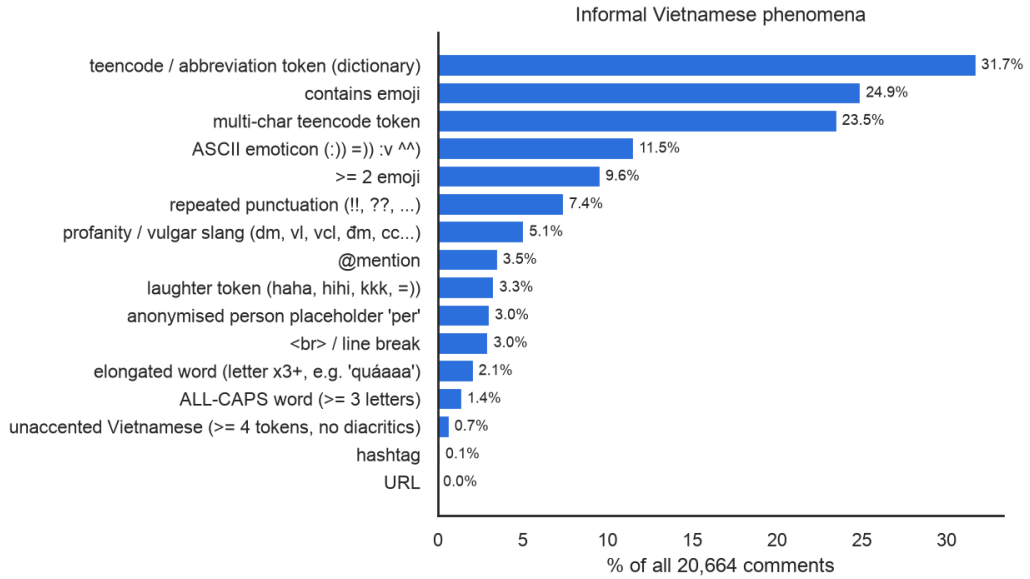
Training labels only · correlation_matrix_train.csv

TEXT LENGTH AND INFORMAL VIETNAMESE

Short, noisy text: a third uses teencode, a quarter uses emoji



Median 11 tokens / 47 characters; p99 = 84 tokens. PhoBERT: p99 = 85 sub-words, so max_len 192 truncates 0.01%.



Emoji are strongest for love (48.8%), pride (44.4%) and amusement (40.8%): emoji must be kept or translated, never deleted.

text_length_stats.csv · informal_text_audit.csv · emoji_rate_by_label.csv (aggregates only)

TRAIN / VALIDATION / TEST PROTOCOL

One fixed split for every model, audited for stratification

Split protocol (80/10/10)	Label-dist. LD ↓	Min val pos.	Min test pos.
Official release (used)	**0.0035**	**58**	**60**
Random, seed 42	0.0044	73	88
Iterative stratified, seed 42	0.0004	75	76

We keep the official split so every number is comparable with the published baselines. The audit shows it is only random-level stratified; the iterative split (Sechidis et al., 2011) is saved for a robustness check in the Final.

Train · 16,531

fit models, vocabularies, co-occurrence statistics

Validation · 2,066

early stopping, hyper-parameters, thresholds

Test · 2,067 (sealed)

scored once per frozen model

splits/official/*_ids.txt · results/tables/split_quality_audit.csv

PREPROCESSING PIPELINE

Normalise the noise, keep the emotion signal

1 · Unicode NFC

fixes a bug in the official code: “cảm” → “cảmày”

2 · Mask & clean

@user, url,
, lower-case

3 · Tone marks

VinAI table: “hòa” → “hoà”, “thủy” → “thuỷ”

4 · Elongation

“quáaaa” → “quá”, “kkk” → “haha”, “:)))))” → “:))”

5 · Emoji

keep as tokens, or 😭 → “khóc lớn” (CLDR vi fallback)

6 · Teencode

token-level lookup, 441 official + 62 added: “ko” → “không”

7 · Word segmentation

VnCoreNLP RDRSegmenter (PhoBERT's own; paper: pyvi)

Cached & hashed

every step is a config switch; cache key = config + code

Model	Emoji	Segmentation	Tokeniser
TF-IDF + LR, BiLSTM	keep as tokens	VnCoreNLP	word + char n-grams / train vocabulary
PhoBERT	→ Vietnamese text	VnCoreNLP	PhoBERT BPE (64k)
ViSoBERT	keep (in vocabulary)	none (syllables)	SentencePiece (15k)

src/vigo/data/preprocessing.py · resources/ (dictionaries credited to Hung et al. 2026, VinAI, Unicode CLDR)

MODELS

From sparse n-grams to Vietnamese transformers

Model	Architecture	Training
M1 · TF-IDF + OvR LR	word 1–2-grams + char 1–5-grams (76k features), 28 logistic regressions	C and class weight chosen on val (C = 4, balanced)
M2 · BiLSTM	300-d embeddings (from scratch), 2 × BiLSTM 256, attention + max pooling	BCE, AdamW one-cycle 2e-3, early stop on val
M3 · PhoBERT	vinai/phobert-base-v2, CLS → dropout → 28 sigmoids	BCE, lr 3e-5, layer-wise decay 0.9, ≤ 10 epochs
+ ViSoBERT	uitnlp/visobert: social-media RoBERTa, emoji in vocabulary	same recipe, lr 4e-5
+ ViSoBERT-14GB	5CD-AI continued pre-training on 14 GB of Facebook comments	same recipe, lr 2e-5
+ Ensemble	mean of the saved probabilities of the 3 transformers	no retraining, no test tuning

Shared tricks: label-prior bias initialisation, length-grouped dynamic padding, best epoch chosen on validation Macro-F1 at τ = 0.5. CafeBERT and PhoBERT-large configs are ready for a larger GPU (not run).

METRICS AND DECODING PROTOCOL

One evaluation module, one threshold policy, for all models

Required metrics

- Macro-F1 and Micro-F1 (headline)
- Per-label F1 for all 28 labels
- Hamming loss (lower is better)
- LRAP and macro mAP (threshold-free ranking)
- Parameter count, wall-clock training time

Decoding

- Main table: fixed $\tau = 0.5$ for every label
- Probabilities are saved before thresholding
- Global τ^* and per-label τ^*_c are fitted on validation only, frozen, then applied once to test
- No subset accuracy as a headline metric

src/vigo/evaluation/ (metrics, thresholds) · src/vigo/analysis/midterm.py

MAIN RESULTS · TEST SET, T = 0.5

PhoBERT reaches 0.629 Macro-F1; the ensemble reaches 0.643

Model	Macro-F1	Micro-F1	Hamming ↓	LRAP	mAP	Params	Train time
M1 · TF-IDF + OvR LR	0.542	0.558	0.0665	0.690	0.560	2.1M	6 s
M2 · BiLSTM	0.512	0.533	0.0581	0.668	0.539	5.2M	21.6 min
M3 · PhoBERT-base-v2	**0.629**	**0.646**	**0.0472**	**0.769**	**0.684**	**134.4M**	**51.0 min**
ViSoBERT	0.613	0.632	0.0483	0.755	0.656	97.0M	41.1 min
ViSoBERT-14GB	0.620	0.642	0.0470	0.777	0.678	97.0M	41.9 min
Ensemble of the 3 transformers	**0.643**	**0.664**	**0.0429**	**0.793**	**0.710**	**328.4M**	**134.0 min**

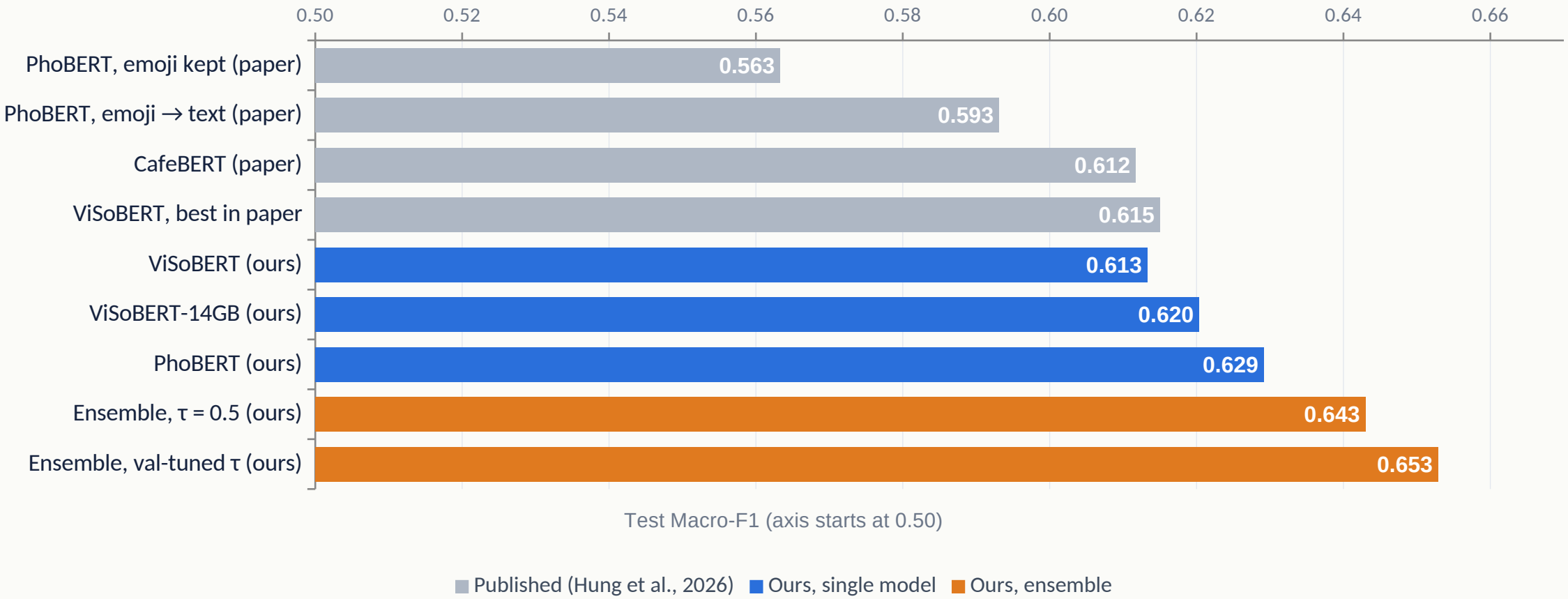
PhoBERT beats the recurrent baseline by **+11.7** Macro-F1 points and TF-IDF by **+8.7**.

With a global $\tau^* = 0.32$ fitted on val, the ensemble reaches **0.653 Macro-F1 / 0.669 Micro-F1**.

Apple M1 Pro 32 GB (MPS; BiLSTM on CPU), seed 42 · main_results_test.csv, threshold_strategies.csv, per_label_f1.csv

COMPARISON WITH PUBLISHED BASELINES · SAME OFFICIAL SPLIT

PhoBERT alone beats the paper's best model

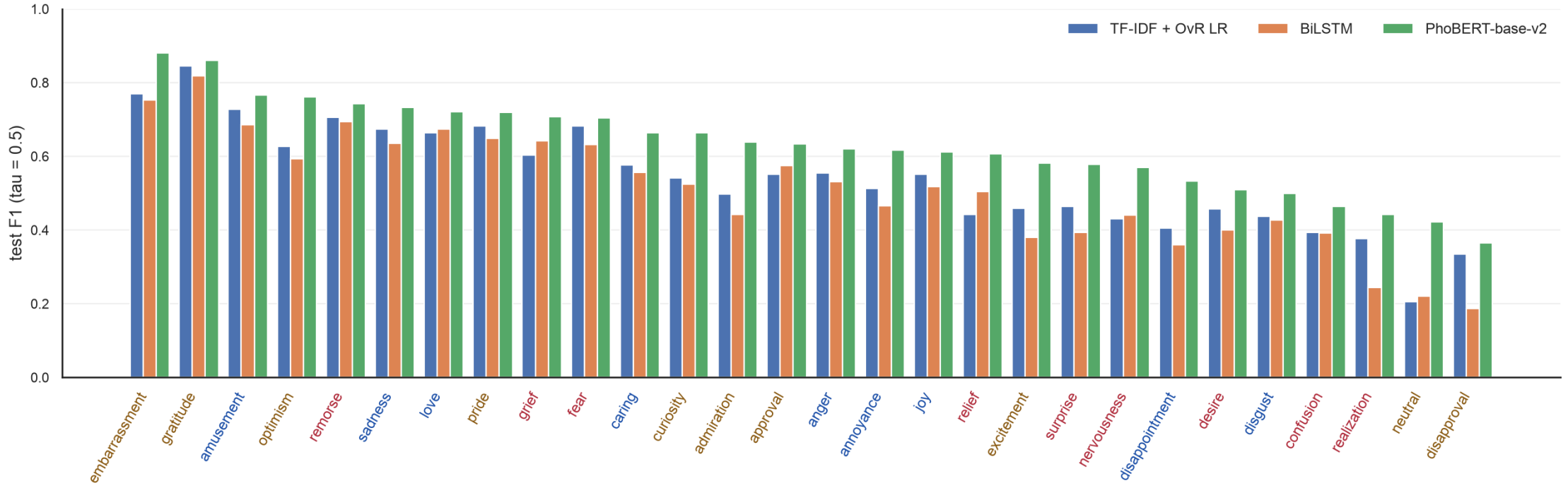


What differs from the paper's recipe (not yet ablated one by one): NFC and tone normalisation, VnCoreNLP segmentation, extended emoji and teencode dictionaries, plain BCE with prior-initialised bias, layer-wise LR decay at 3e-5 instead of 5e-5.

Paper numbers: Hung et al. (EACL 2026) official notebooks, τ = 0.5, pos-weighted BCE · comparison_with_paper.csv

Q1 · HIGHEST AND LOWEST PER-LABEL F1

All three models agree on the easy and the hard labels



Highest (PhoBERT)

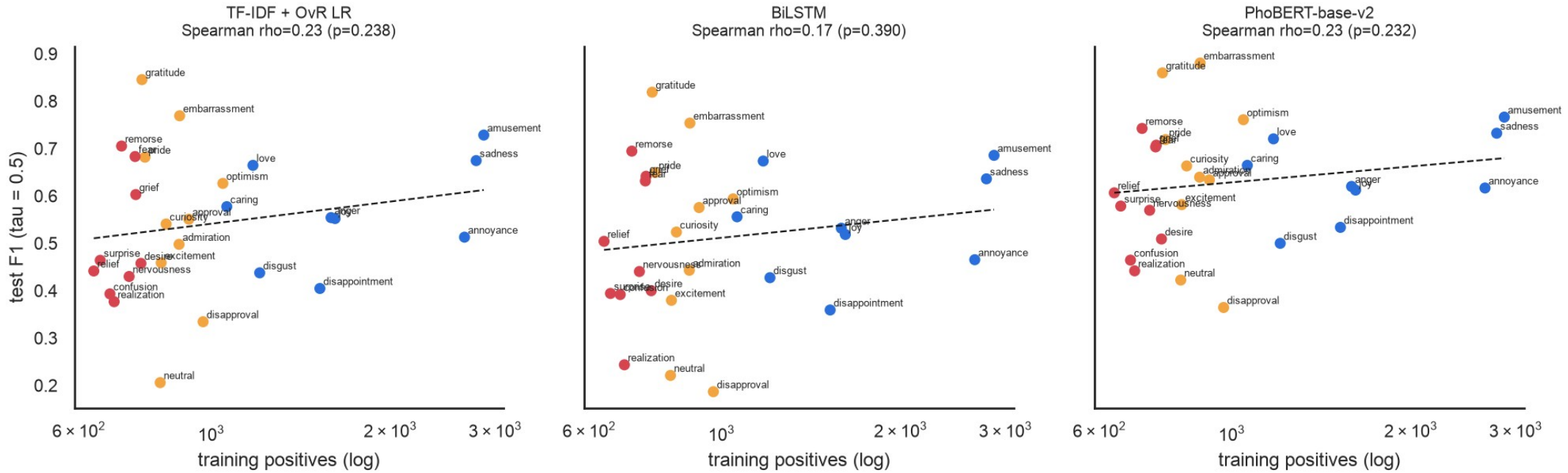
embarrassment 0.88 · gratitude 0.86 · amusement 0.77 · optimism 0.76 · remorse 0.74

Lowest (PhoBERT)

disapproval 0.36 · neutral 0.42 · realization 0.44 · confusion 0.46 · disgust 0.50

Test set, τ = 0.5 · tick colour = tier (blue head, brown mid, red tail) · per_label_f1.csv, q1_best_worst_labels.csv

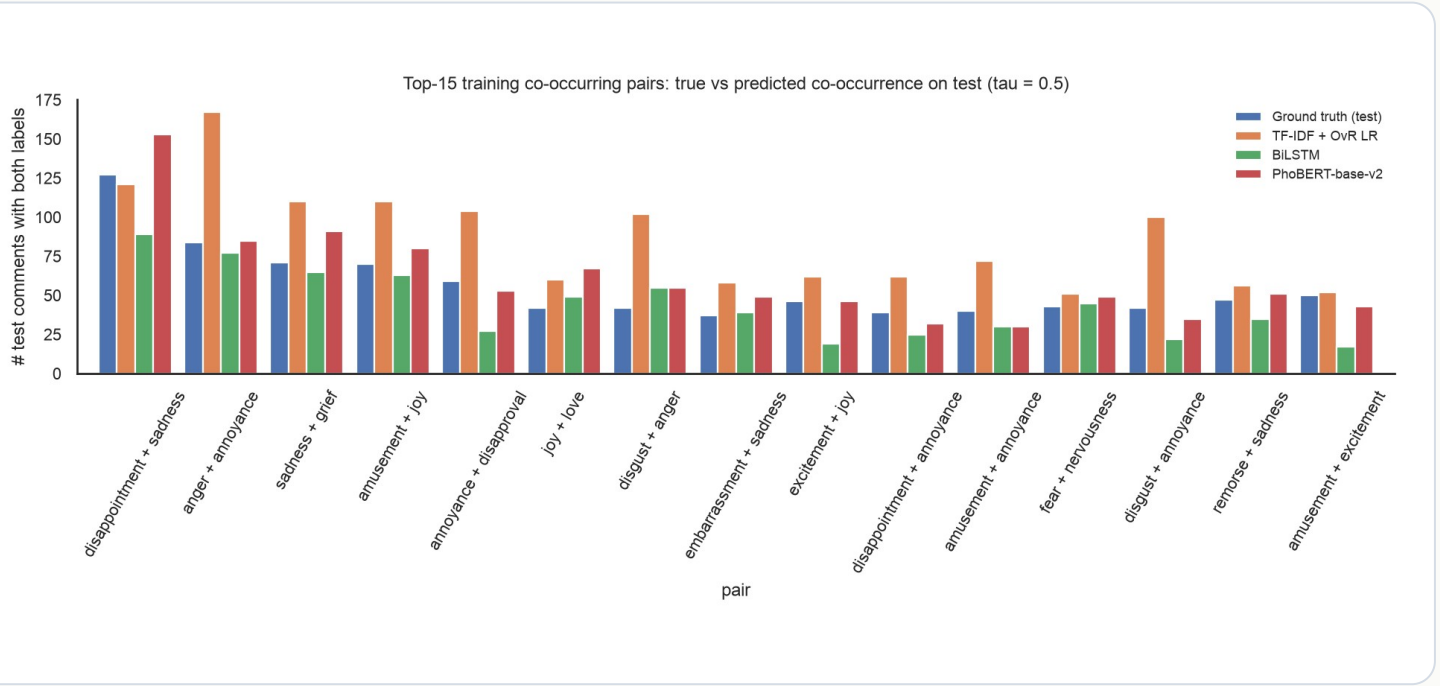
Frequency barely predicts F1: difficulty is semantic



Model	Spearman ρ (p)	Head F1	Mid F1	Tail F1
TF-IDF + LR	0.23 (0.24)	0.567	0.551	0.506
BiLSTM	0.17 (0.39)	0.539	0.515	0.482
PhoBERT	0.23 (0.23)	0.641	0.653	0.592

28 labels, test F1 at τ = 0.5 vs training positives · q2_frequency_correlation.csv · correlation ≠ causation

PhoBERT reproduces the pair structure; the baselines distort it



Model	r (co-occ.)	Pred. |Y|
Ground truth	–	1.91
TF-IDF + LR	0.921	2.30
BiLSTM	0.940	1.58
PhoBERT	**0.972**	**1.83**

TF-IDF over-pairs

4,698 predicted pairs vs 2,466 true; disgust + anger 2.4×, anger + annoyance 2.0×.

BiLSTM under-pairs

1,658 pairs; excitement + joy 0.41×, annoyance + disapproval 0.46×.

PhoBERT

2,314 pairs, 0.75–1.6× per pair; over-predicts joy + love (1.6×).

Top pairs from TRAIN labels; r = Pearson between true and predicted pair counts over all 378 pairs on test · q3_*.csv

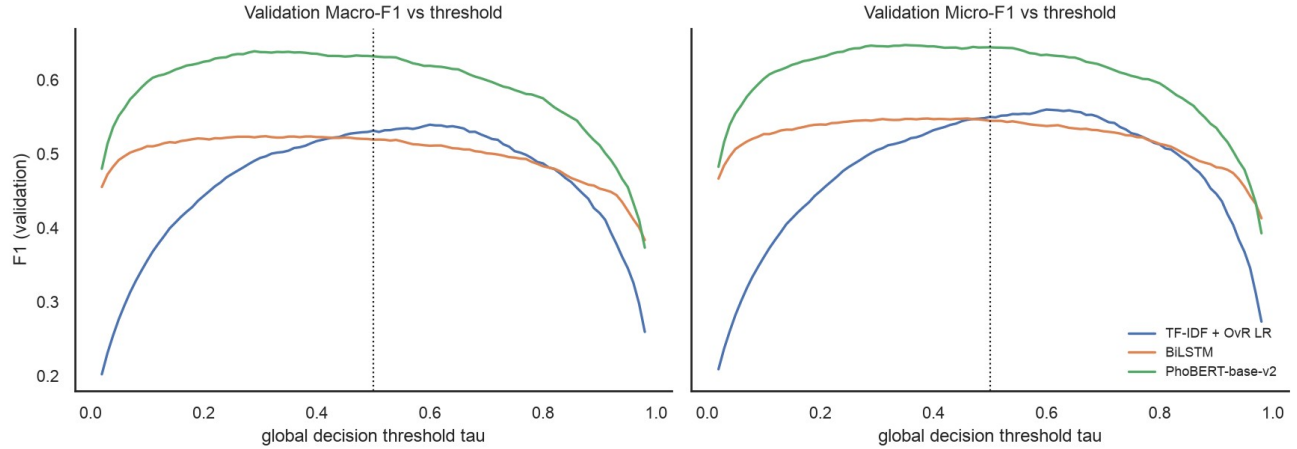
Biggest gains on emoticon, long and teencode text

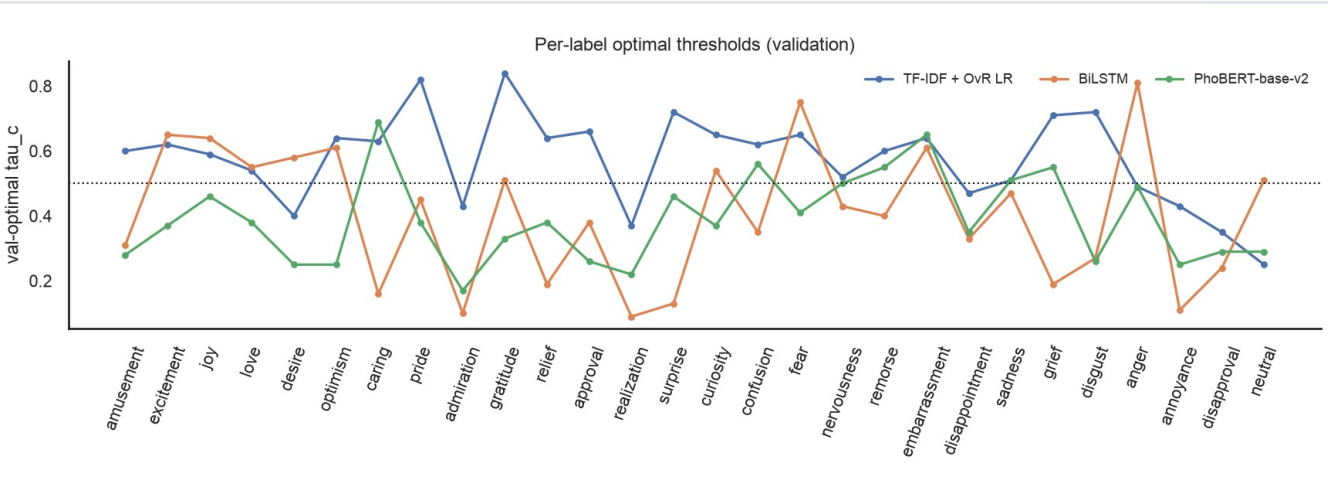
Subset (rule frozen before evaluation)	n	BiLSTM	PhoBERT	Δ Macro	Δ Micro, 95% CI
emoticon (≥ 1 ":))", "=))", ":v")	**236**	**0.419**	**0.608**	**+0.189**	**[+0.099, +0.179]**
long (≥ 30 tokens)	**283**	**0.414**	**0.577**	**+0.163**	**[+0.099, +0.172]**
plain (no style marker)	793	0.486	0.628	+0.142	[+0.128, +0.174]
teencode (≥ 1 abbreviation)	**509**	**0.468**	**0.596**	**+0.128**	**[+0.099, +0.155]**
all test comments	2,067	0.512	0.629	+0.117	[+0.098, +0.128]
short (≤ 5 tokens)	337	0.553	0.658	+0.104	[+0.042, +0.108]
emoji-heavy (≥ 2 emoji)	180	0.531	0.619	+0.088	[+0.024, +0.105]
emoji (≥ 1)	536	0.531	0.590	+0.059	[+0.019, +0.077]

Emoji are a strong lexical cue that the BiLSTM learns directly as tokens, so pre-trained context adds least there. Emoticons, slang and long comments need context — PhoBERT gains most there.

Test Macro-F1 at τ = 0.5 · CI = paired bootstrap (500 resamples) of the Micro-F1 difference · q4_text_style_breakdown.csv

0.5 is close to optimal on average, but not for each label



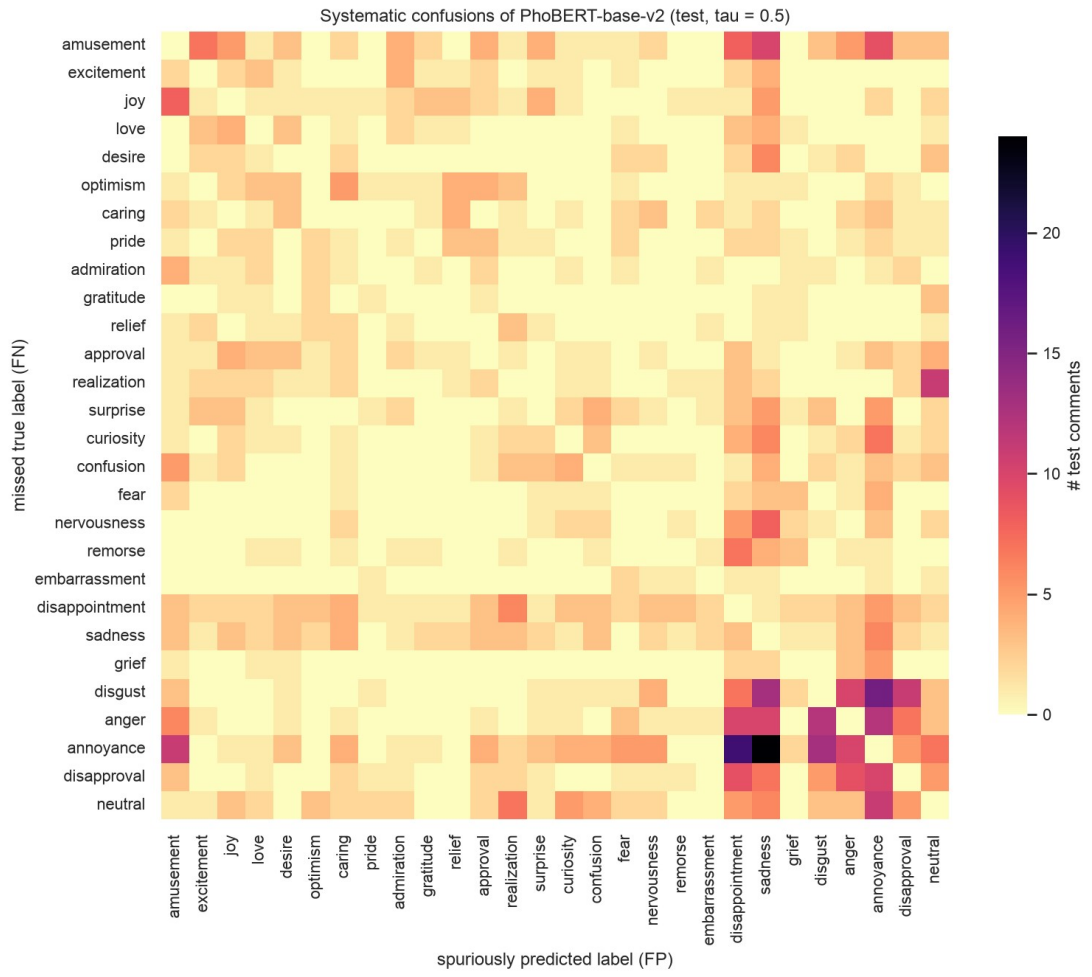


Test Macro-F1	τ = 0.5	global τ*	per-label
TF-IDF	0.542	0.538	0.539
BiLSTM	0.512	0.524	0.518
PhoBERT	**0.629**	**0.633**	**0.634**

PhoBERT: val-optimal τ*_c spans **0.17** (admiration) to **0.69** (caring), median 0.38.

Per-label tuning gains +0.023 on val but only +0.004 on test: a sign of threshold overfitting.

Thresholds fitted on validation only, then frozen · threshold_strategies.csv · q5_*.csv



ERROR ANALYSIS · PHOBERT, TEST

Four recurring failure modes

1 · Confusion inside a family

annoyance missed while sadness (24) or disappointment (19) is predicted; disgust ↔ annoyance ↔ anger.

2 · Implicit, context-bound labels

Miss rates: disapproval 68.8%, neutral 61.2%, realization 60.0%, confusion 57.1%.

3 · Pull towards two labels

$|Y| = 1$: 0.76 spurious labels per comment. $|Y| \geq 4$: 1.89 labels missed.

4 · Long comments

Exact match 19.8% vs 44.8% for short ones; 0.89 vs 0.45 missed labels per comment.

τ = 0.5 · error_*.csv; 97 cases for manual audit (25 FN, 25 FP, 15 multi-label, 32 by style)

PROPOSED FINAL RESEARCH QUESTION

RQ1: Is a single uniform threshold ($\tau = 0.5$) optimal across all 27 emotions under class imbalance?

Midterm evidence

Optimal τ^*_c ranges 0.17–0.69. A lower global τ helps $|Y| \geq 4$ (0.67 → 0.75 Micro-F1) but hurts $|Y| = 1$ (0.57 → 0.53). The val gain of per-label tuning shrinks on test.

Controlled design

Best Midterm network, backbone frozen. Compare fixed, global and per-label τ fitted on val. Break down by head/mid/tail and $|Y| = 1, 2, 3, \geq 4$; report the val→test gap.

Robustness

3 seeds with bootstrap CIs, per-label τ shrunk towards the global one, and a check on the iterative-stratified split.

Optional extensions: label-dependency modelling (RQ2) motivated by Q3 and the family confusions; imbalance-aware losses (RQ3); noise robustness on teencode and unaccented text (RQ4).

Probabilities for val and test are already saved in the Final schema: results/predictions/

SUMMARY

Four take-aways from the Midterm

1 · Strong, valid baselines

PhoBERT reaches 0.629 Macro-F1 and the ensemble 0.653, against 0.615 for the best published model — same split, test sealed.

2 · Ambiguity beats rarity

Frequency explains little ($\rho \approx 0.2$, n.s.); disapproval, neutral and realization are hard because they are implicit.

3 · Context helps where text is noisy

Largest gains on emoticon, long and teencode comments; errors stay inside emotion families.

4 · One threshold does not fit all

Optimal τ^*_c ranges 0.17–0.69, but per-label tuning overfits validation — the Final RQ1.

Reproducible end to end: **`bash scripts/run_all.sh`** or the Colab notebook — fixed split ids, YAML configs, seed 42, saved probabilities.